

Image Generation Models

Project: Verified Study Buddy

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Prompt Evaluation

Without opening the old lesson, explain one key idea from Prompt Evaluation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Iterative Prompting

Without opening the old lesson, explain one key idea from Iterative Prompting. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Sampling and Variability

Without opening the old lesson, explain one key idea from Sampling and Variability. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Image generation and provenance. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Image generation and provenance. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

A generated diagram requires two arrows and labels input/output. It has two arrows but swapped labels. Metadata records generation, yet the caption says experimental observation. Audit.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

An image generator produces a convincing picture of a school greenhouse on Mars. A student calls it photographic evidence. Explain what the image can and cannot establish.

6. Solve the given case

A generator creates a realistic fictional bicycle beside a tree.

How does this differ from event photography?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

An image model draws a plant with seven roots after a request for exactly four. What does this show, and where should an exact count diagram be made?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 10 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Treat generated diagrams as illustrations; pair factual claims with approved sources.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

An image generator produces a convincing picture of a school greenhouse on Mars. A student calls it photographic evidence. Explain what the image can and cannot establish.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

13. W2 Trace the worked case

A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

14. W3 Audit the limit

A report omits this qualification: Provenance tracks production history; it does not prove the content is accurate or establish every legal permission. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

15. W4 Apply a new case independently

A generated diagram requires two arrows and labels input/output. It has two arrows but swapped labels. Metadata records generation, yet the caption says experimental observation. Audit.

My reasoning and evidence

16. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Image Generation Models

A fictional lesson poster illustrates one family of image-generation approaches.

Toy 2×2 noisy numerical grid: $[[0.2, 0.8], [0.6, 0.3]]$. A specified illustrative denoising step gives $[[0.1, 0.9], [0.4, 0.2]]$.

These supplied arrays illustrate steps only; they are not outputs from a real trained diffusion model.

Illustrative grids, not executed model outputs

	Left	Right
Before top	0.2	0.8
Before bottom	0.6	0.3
After top	0.1	0.9
After bottom	0.4	0.2

Read a cell using both its row and column labels.

Image Generation Models

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Noisy representation

Four toy numerical cells form the starting grid.



2. Condition on request

A text request guides the imagined learned denoising process.



3. Illustrative update

Use the supplied second grid to show one change in representation.



4. Check provenance

Caption it as a generated-process illustration, not a photographed experiment.

CHANGE ONE THING

Relabel the illustration as a photograph of an actual laboratory measurement.

Explain the changed result using the matching lesson.

What may these two grids explain, and what can they not establish about a real generator?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Use Study Buddy generated diagrams only with clear provenance and checked educational labels.

19. Project portfolio milestone

Project: Verified Study Buddy

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

Use Study Buddy generated diagrams only with clear provenance and checked educational labels.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional lesson poster illustrates one family of image-generation approaches.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Caption it as a generated-process illustration, not a photographed experiment.

1. Noisy representation
2. Condition on request
3. Check provenance
4. Illustrative update

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Check provenance Caption it as a generated-process illustration, not a photographed experiment. Diffusion-style generators learn to remove noise in stages, often guided by text, but a real implementation can operate in a latent representation rather than direct pixels. The toy values let us inspect an update without pretending to simulate that trained process. An attractive final image still needs checks for labels, requested relations and factual claims.

Practice 2: explain the deciding evidence

Changed-case practice: Relabel the illustration as a photograph of an actual laboratory measurement. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. That caption is unsupported.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: That caption is unsupported. Its provenance is an invented diagram of a process, not a camera record of an event.

Repair a missing building block

Review Prompt Evaluation (ai-3/9). Goal: Explain the mechanism and terminology of controlled prompt experiments. Short practice: Why is B's selection defensible here without calling it universally superior?. Return to this week and explain what you changed.

Review Iterative Prompting (ai-3/7). Goal: Explain the mechanism and terminology of controlled prompt experiments. Short practice: What evidence supports improvement here, and what remains unproven?. Return to this week and explain what you changed.

Review Sampling and Variability (ai-3/4). Goal: Sample from a distribution using explicit cumulative intervals. Short practice: Does $T=0.5$ make B impossible, and does choosing A guarantee factual accuracy?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
